

The Solar System

The Sun and Its Family

The Solar System is centered on the Sun, a star whose gravity holds planets, dwarf planets, moons, asteroids, and comets in orbit. The eight planets are Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune. They differ greatly in size, temperature, composition, atmosphere, and distance from the Sun.

The inner planets—Mercury, Venus, Earth, and Mars—are rocky worlds. Jupiter and Saturn are gas giants, while Uranus and Neptune are ice giants. The planets follow elliptical paths around the Sun, although most planetary orbits are close to circular.

Mercury

Mercury is the smallest planet and the closest planet to the Sun. Its surface is covered with craters and large rocky plains. Mercury has almost no substantial atmosphere, so temperatures can change dramatically between its day and night sides.

Venus

Venus is the second planet from the Sun and is similar to Earth in size. Its atmosphere is extremely thick and is dominated by carbon dioxide. Thick clouds surround the planet, and the strong greenhouse effect makes Venus the hottest planet in the Solar System.

Earth

Earth is the third planet from the Sun. It has abundant liquid water, a nitrogen-rich atmosphere, and conditions that support a wide variety of life. Earth has one natural satellite, the Moon, which orbits our planet and influences ocean tides.

Mars

Mars is the fourth planet from the Sun and is commonly called the Red Planet because iron-bearing minerals in its surface material give it a reddish appearance. Mars has polar ice caps, volcanoes, valleys, and evidence of ancient water activity.

Jupiter

Jupiter is the fifth planet from the Sun and the largest planet in the Solar System. It is a rapidly rotating gas giant with a thick atmosphere. Jupiter is famous for the Great Red Spot, a huge storm that has been observed for centuries.

Saturn

Saturn is the sixth planet from the Sun. It is a gas giant surrounded by a broad and bright ring system made mainly of particles of ice and rock. Saturn has many moons, including Titan,

which has a substantial atmosphere.

Uranus

Uranus is the seventh planet from the Sun. It is an ice giant with an atmosphere containing hydrogen and helium along with methane. Uranus has a distinctive axial tilt, causing it to experience unusual seasonal changes during its long orbit around the Sun.

Neptune

Neptune is the eighth planet from the Sun and the most distant of the eight recognized planets. It is a cold ice giant with a dynamic atmosphere and extremely fast winds. Neptune appears blue because methane in its atmosphere absorbs much of the red light.

The Moon

The Moon is Earth's natural satellite. It does not produce its own visible light; instead, its surface reflects sunlight. The Moon's gravitational interaction with Earth contributes strongly to ocean tides, and its phases result from changing illumination as it moves around Earth.

Asteroids and Comets

Asteroids are rocky or metallic objects that orbit the Sun. Many are found in the asteroid belt between Mars and Jupiter. Comets contain ice, dust, and rocky material. When a comet approaches the Sun, heat can cause material to escape from its surface and form a glowing coma and, under suitable conditions, a tail.

The Outer Solar System

Beyond Neptune lies a broad region containing many small icy bodies. The Kuiper Belt is one important region of the outer Solar System. Farther away, the Oort Cloud is a proposed distant reservoir of icy objects and is associated with the origin of some long-period comets.

Planetary Order

Starting from the Sun, the planets occur in this order: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune. This order is useful when comparing planetary distances and understanding the overall structure of the Solar System.

Quick Facts

The largest planet is Jupiter. The smallest planet is Mercury. Earth is the third planet from the Sun. Mars is the fourth planet. Saturn is well known for its rings. Neptune is the eighth planet. Earth's natural satellite is the Moon.